

CO223_LAB03

E/21/019

1. Run nslookup to obtain the IP address of a Web server in Asia. What is the IP address of that server?

```
> nslookup www.slt.lk
DNS request timed out.
    timeout was 2 seconds.
Server:  www.slt.lk
Address:  220.247.222.74

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to www.slt.lk timed-out
```

2. Run nslookup to determine the authoritative DNS servers for a university in Europe.

```
C:\Users\hirus>nslookup -type=NS www.cam.ac.uk
Server:  UnKnown
Address:  fe80::1

cam.ac.uk
    primary name server = primary.dns.cam.ac.uk
    responsible mail addr = hostmaster.cam.ac.uk
    serial      = 1734978646
    refresh     = 1800 (30 mins)
    retry       = 900 (15 mins)
    expire      = 604800 (7 days)
    default TTL = 3600 (1 hour)

C:\Users\hirus>
```

3. Run nslookup so that one of the DNS servers obtained in Question 2 is queried for the mail servers for Yahoo! mail. What is its IP address?

- IP Address is 2406:2000:e4:1604::1001

```

C:\Users\hirus>nslookup primary.dns.cam.ac.uk mail.yahoo.com
DNS request timed out.
    timeout was 2 seconds.
Server:    UnKnown
Address:   2406:2000:e4:1604::1001

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to UnKnown timed-out

C:\Users\hirus>

```

4. Locate the DNS query and response messages. Are then sent over UDP or TCP?

- UDP

The image shows a Wireshark packet capture of a DNS query and response. The query (packet 14) is sent from 192.168.1.14 to 192.168.1.1. The response (packet 15) is sent from 192.168.1.1 to 192.168.1.14. The response contains IP addresses for www.ietf.org.

Packet 14: 72 bytes on wire (576 bits), 72 bytes captured (576 bits) on interface \Device\NPF{1B9CEA87-2...}

Ethernet II, Src: Intel_2b:9a:1c (6c:94:66:2b:9a:1c), Dst: HuaweiTechno_a7:d8:a4 (ec:c0:1b:a7:d8:a4)

Internet Protocol Version 4, Src: 192.168.1.14, Dst: 192.168.1.1

User Datagram Protocol, Src Port: 57026, Dst Port: 53

Source Port: 57026

Destination Port: 53

Length: 38

Checksum: 0x8397 [unverified]

[Checksum Status: Unverified]

[Stream index: 1]

[Stream Packet Number: 1]

[Timestamps]

UDP payload (38 bytes)

Domain Name System (query)

5. What is the destination port for the DNS query message? What is the source port of DNS response messages?

```

Source Port: 57026
Destination Port: 53

```

- The destination port is port 53, and the source port is port 57026.

6. To what IP address is the DNS query message sent? Use ipconfig to determine the IP address of your local DNS server. Are these two IP addresses the same?

```

DNS Servers . . . . . : fe80::1%5
                        192.168.1.1

```

```
▶ Internet Protocol Version 4, Src: 192.168.1.14, Dst: 192.168.1.1
▼ User Datagram Protocol, Src Port: 57026, Dst Port: 53
    Source Port: 57026
    Destination Port: 53
    Length: 38
```

- The DNS query message was sent to 192.168.1.1. This is the same IP address as the local DNS server

7. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

```
▼ Domain Name System (query)
    Transaction ID: 0xec2c
    ▶ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
    ▼ Queries
        ▶ www.ietf.org: type AAAA, class IN
```

- This query was a type AAAA query. It did not contain any “answers”.

8. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain?

```
▼ Domain Name System (response)
    Transaction ID: 0xec2c
    ▶ Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 0
    ▼ Queries
        ▼ www.ietf.org: type AAAA, class IN
            Name: www.ietf.org
            [Name Length: 12]
            [Label Count: 3]
            Type: AAAA (28) (IP6 Address)
            Class: IN (0x0001)
        ▼ Answers
            ▶ www.ietf.org: type AAAA, class IN, addr 2606:4700::6810:2d63
            ▶ www.ietf.org: type AAAA, class IN, addr 2606:4700::6810:2c63
            [Request In: 14]
            [Time: 0.011941000 seconds]
```

- This DNS response message provided two answers. The answers contain the address of the website that it was queried for.

▼ Answers

- ▼ www.ietf.org: type AAAA, class IN, addr 2606:4700::6810:2d63
 - Name: www.ietf.org
 - Type: AAAA (28) (IP6 Address)
 - Class: IN (0x0001)
 - Time to live: 108 (1 minute, 48 seconds)
 - Data length: 16
 - AAAA Address: 2606:4700::6810:2d63
- ▼ www.ietf.org: type AAAA, class IN, addr 2606:4700::6810:2c63
 - Name: www.ietf.org
 - Type: AAAA (28) (IP6 Address)
 - Class: IN (0x0001)
 - Time to live: 108 (1 minute, 48 seconds)
 - Data length: 16
 - AAAA Address: 2606:4700::6810:2c63

9. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message?

No.	Time	Source	Destination	Protocol	Length	Info
32	11:41:10.904208	2402:d000:8110:2e11::	2606:4700::6810:2d63	TCP	86	13796 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM
40	11:41:10.916087	2402:d000:8110:2e11::	2606:4700::6810:2d63	TCP	86	13797 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM
54	11:41:10.941091	2402:d000:8110:2e11::	64:ff9b::284a:51c6	TCP	86	13798 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM
56	11:41:10.941385	2402:d000:8110:2e11::	64:ff9b::284a:51c6	TCP	86	13799 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM
75	11:41:11.074440	2402:d000:8110:2e11::	64:ff9b::284a:51c6	TCP	86	13800 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM
244	11:41:11.577692	2402:d000:8110:2e11::	2606:4700::6810:2c63	TCP	86	13801 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM
260	11:41:11.589709	2402:d000:8110:2e11::	2606:4700::6810:2c63	TCP	86	13802 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1412 WS=256 SACK_PERM

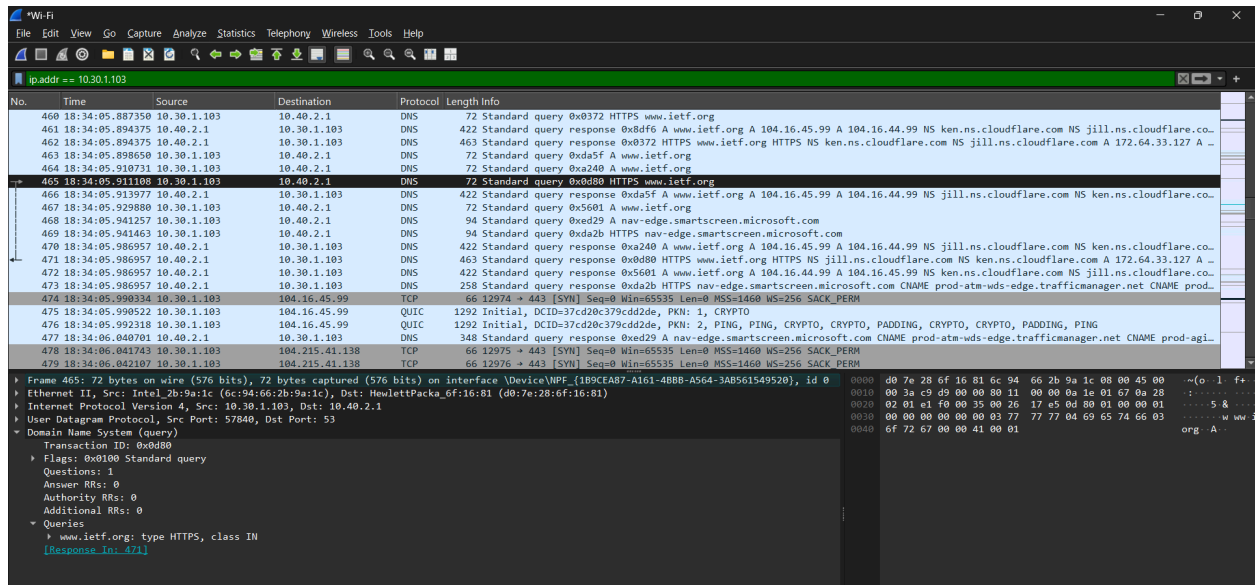
Frame 32: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF{1B9CEA87-A161-4B8B-A564-3AB561549520}, id 0
 Ethernet II, Src: Intel_2b:9a:1c (6c:94:66:2b:9a:1c), Dst: HuaweiTechno_a7:d8:a4 (ec:c0:1b:a7:d8:a4)
 Internet Protocol Version 6, Src: 2402:d000:8110:2e11:786d:202a:ddd:b73, Dst: 2606:4700::6810:2d63
 0110 = Version: 6
 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
 1110 1011 1111 0000 1010 = Flow Label: 0xebf0a
 Payload Length: 32
 Next Header: TCP (6)
 Hop Limit: 64
 Source Address: 2402:d000:8110:2e11:786d:202a:ddd:b73
 Destination Address: 2606:4700::6810:2d63
 [Stream index: 5]
 Transmission Control Protocol, Src Port: 13796, Dst Port: 443, Seq: 0, Len: 0

0000 ec c0 1b a7 d8 a4 6c 94 66 2b 9a 1c 8
 0010 bf 0a 00 20 06 40 24 02 d0 00 81 10 2
 0020 20 2a 0d dd 0b 73 26 06 47 00 00 00 0
 0030 00 00 68 10 2d 63 35 e4 01 bb 69 2a 0
 0040 00 00 80 02 ff ff 57 ac 00 00 02 04 0
 0050 03 00 01 01 04 02

- Addresses are the same.

10. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

- ▼ Queries
 - ▶ www.ietf.org: type HTTPS, class IN
[Response In: 471]



11. What is the destination port for the DNS query message? What is the source port of DNS response message?

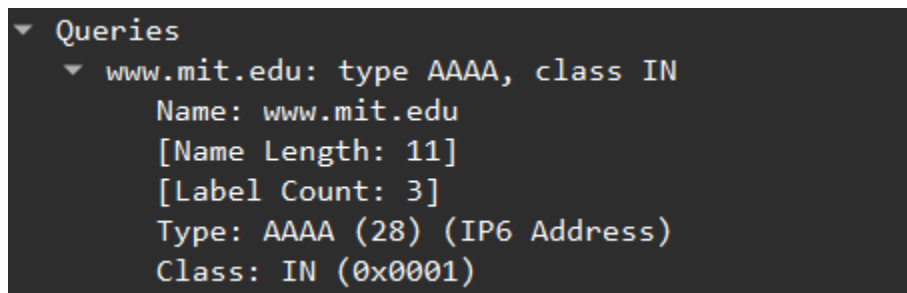
Source Port: 60287
 Destination Port: 53

12. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

Internet Protocol Version 4, Src: 192.168.1.14, Dst: 192.168.1.1

DNS Servers : fe80::1%5
 192.168.1.1

13. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?



- Type is AAAA. They have 4 answers.

14. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain?

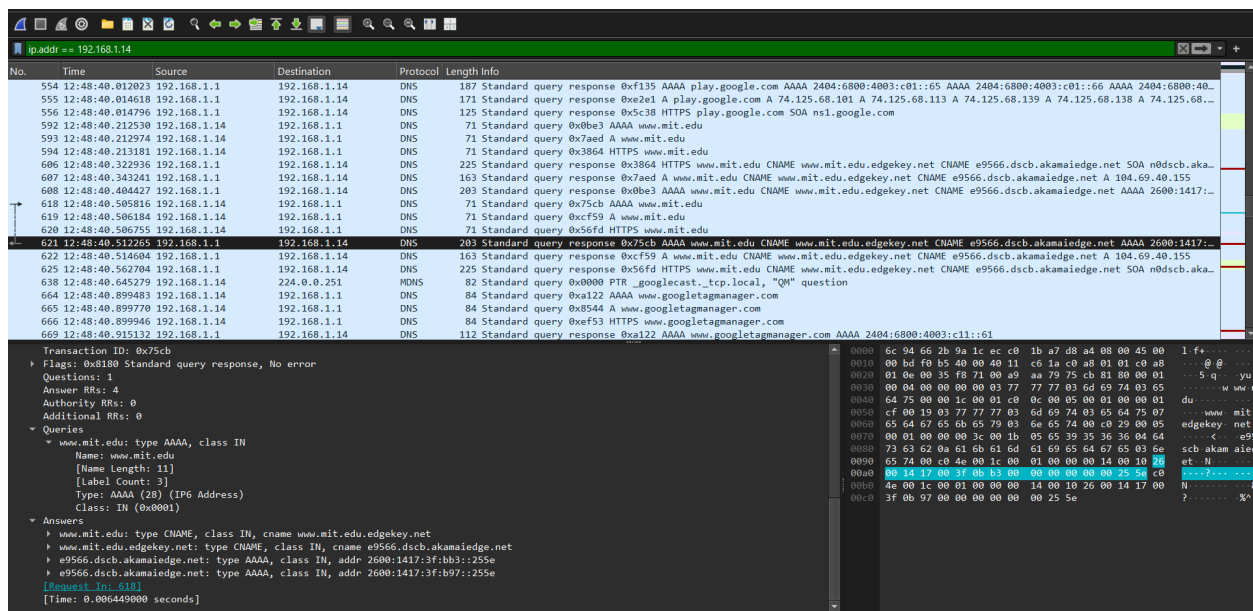
```

Queries
  www.mit.edu: type AAAA, class IN
    Name: www.mit.edu
    [Name Length: 11]
    [Label Count: 3]
    Type: AAAA (28) (IP6 Address)
    Class: IN (0x0001)

Answers
  www.mit.edu: type CNAME, class IN, cname www.mit.edu.edgekey.net
  www.mit.edu.edgekey.net: type CNAME, class IN, cname e9566.dscb.akamaiedge.net
  e9566.dscb.akamaiedge.net: type AAAA, class IN, addr 2600:1417:3f:bb3::255e
  e9566.dscb.akamaiedge.net: type AAAA, class IN, addr 2600:1417:3f:b97::255e

```

15. Provide a screenshot.



16. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

```

Internet Protocol Version 4, Src: 192.168.1.14, Dst: 192.168.1.1
...
DNS Servers . . . . . : fe80::1%5
                        192.168.1.1

```

17. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

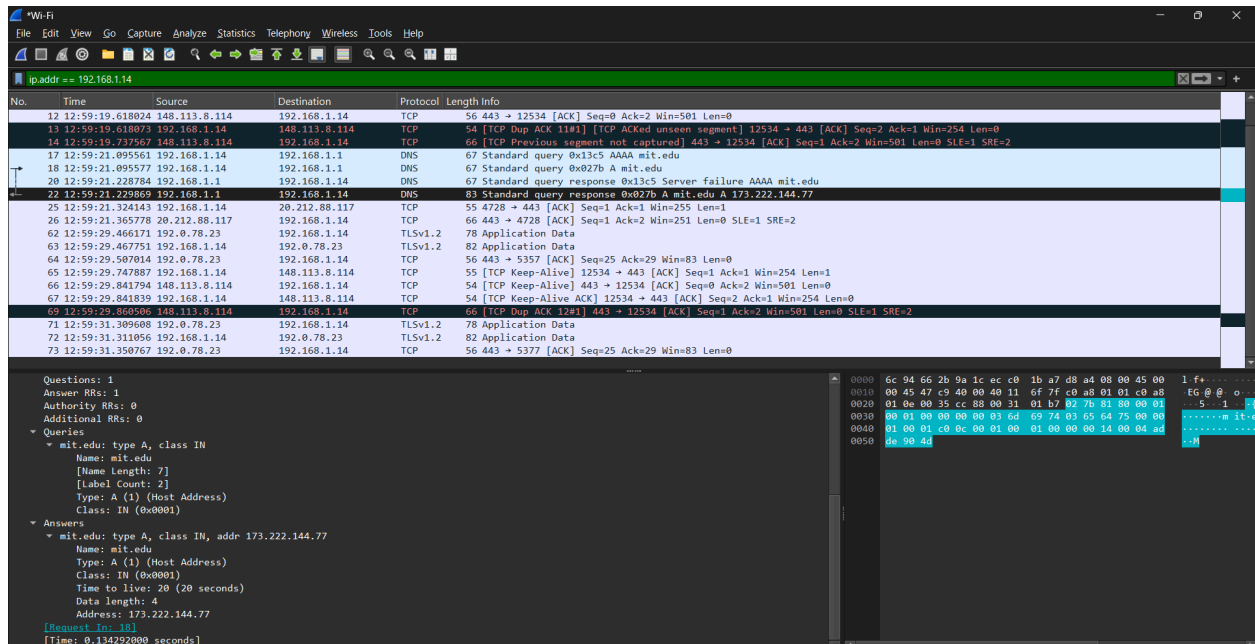
```
Domain Name System (query)
  Transaction ID: 0x13c5
  ▸ Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  ▾ Queries
    ▾ mit.edu: type AAAA, class IN
      Name: mit.edu
      [Name Length: 7]
      [Label Count: 2]
      Type: AAAA (28) (IP6 Address)
      Class: IN (0x0001)
[Response Truncated]
```

- one answer

18. Examine the DNS response message. What MIT nameservers does the response message provide? Does this response message also provide the IP addresses of the MIT nameservers?

```
Domain Name System (response)
  Transaction ID: 0x027b
  ▸ Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 1
  Authority RRs: 0
  Additional RRs: 0
  ▾ Queries
    ▾ mit.edu: type A, class IN
      Name: mit.edu
      [Name Length: 7]
      [Label Count: 2]
      Type: A (1) (Host Address)
      Class: IN (0x0001)
  ▾ Answers
    ▾ mit.edu: type A, class IN, addr 173.222.144.77
      Name: mit.edu
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 20 (20 seconds)
      Data length: 4
      Address: 173.222.144.77
```

19. Provide a screenshot.

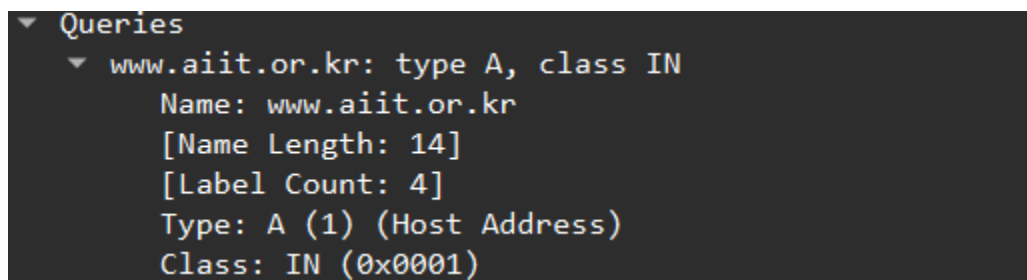


20. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? If not, what does the IP address correspond to?

155	13:06:07.656826	192.168.1.14	18.0.72.3	DNS	74	Standard query 0x0003 AAAA www.aiit.or.kr
156	13:06:07.659678	192.168.1.14	18.0.72.3	TCP	56	443 → 6119 [ACK] Seq=26 Ack=2 Win=63 Len=0
166	13:06:09.662434	192.168.1.14	18.0.72.3	DNS	74	Standard query 0x0004 A www.aiit.or.kr

- The DNS query message is sent to 18.72.0.3 which is not the same as my local DNS server. This IP address corresponds to www.aiit.or.kr.

21. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?



- No answers.

22. Examine the DNS response message. How many “answers” are provided? What does each of these answers contain?

- No responses.

23. Provide a screenshot.

The screenshot displays the Wireshark network traffic analysis tool. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for file operations, capture control, and analysis. The status bar at the top indicates the IP address filter is set to 'ip.addr == 192.168.1.14'.

The main packet list pane shows a series of network packets. The selected packet is a DNS query (No. 166) from 192.168.1.14 to 18.0.72.3. The packet details pane shows the following structure:

- Domain Name System (Query)
 - Transaction ID: 0x0004
 - Flags: 0x0100 Standard query
 - Questions: 1
 - www.aiit.or.kr: type A, class IN
 - Name: www.aiit.or.kr
 - Name Length: 14
 - Label Count: 4
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)

The packet bytes pane shows the raw data of the DNS query, including the transaction ID (0x0004) and the query flags (0x0100).